

Roe V. Wade Twitter Analysis

BSTT 426 Machine Learning Using Python

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1 Introduction

In 1973, the United States Supreme Court ruled against the existing legislation banning abortion in *Roe v. Wade*, and effectively legalized abortion across the United States. This ruling supported a person's right to abortion in all states until it was overturned in June 2022. At that point in time, trigger laws and new legislation quickly banned abortion access in many states.

While abortion is considered a medical necessity in western medicine, the procedure itself has been met with contentious public opinions in the United States even while the procedure was supported by *Roe v. Wade*. After the most recent supreme court ruling abortion access became a more pressing political topic, and people expressed their opinions and concerns online via Twitter.

1.1 Objective

The goal of this analysis will be to look into the topics that people are discussing relating to the *Roe v. Wade* Supreme Court decision on Twitter. These topics will show how people interpret the outcome of this Supreme Court ruling and the general sentiment of these topics in order to inform further policy and research.

2 Data Description

The dataset we are utilizing in this project contains tweets about *Roe v. Wade*, from the *Roe v. Wade* overturn in June 22 through August 2022 from Kaggle. The data includes many variables, including the location (LOCATION), number of accounts followed for the tweeting account (FOLLOWING), number of accounts followed by the tweeting account (FOLLOWERS), total number of tweets for the account (TOTALTWEETS), the date the account was created (USER-CREATEDDT), the date the tweet was created (TWEETCREATEDDT), the number of retweets (RETWEETCOUNT), the text of the tweet (TEXT), the hashtags used in the tweet (HASHTAGS), the language in the tweet (LANGUAGE), the number of favorites on the tweet (FAVORITECOUNT), and if the tweet was retweeted (ISRETWEET). Our analyses will focus on the TEXT variable for text analyses, as this feature includes all text data for each tweet.

3 Methodology

Preprocessing: The variable TEXT is preprocessed to make the methods easier to interpret and use. The preprocessing steps included making all text lower case, removing references to HTML, removing all punctuation, removing all URLs, tokenizing words, filtering out stopwords, stemming the words, and lemmatizing the words. This results in a set of documents that only include useful words for analysis that have been shortened to their base form. This was done using the nltk package in python.

Bag of Words: The bag of words model is a process of representing text as a form of data that can be used for additional analysis. This is done by utilizing the preprocessed data, specifically the stemmed, lemmatized, and stop-words removed version of the documents, in order to describe the occurrence of the words within the document. The bag of words process will make a list of all of the words and create document vectors by tallying the number of words found in each document. This is done through the creation of a term document matrix, which can then be utilized in further machine learning algorithms.

Term Frequency - Inverse Document Frequency (TF-IDF): TF-IDF measures how relevant a word is with respect to the document it is contained in and the collection of documents. There is a collection of documents called a corpus. The TF in TF-IDF is for term frequency the number of times the word occurs in documents. The DF is document frequency which measures the importance of the document within the corpus so essentially the count of occurrences of term in the set of documents. So DF is the number of documents where the term t is present. IDF in TF-IDF is for Inverse Document Frequency, which measures the informativeness of the term t . All encompassing TF-IDF is a valid measure to evaluate how important a word is to the document in a collection or a corpus.

K-Means Clustering: Clustering is an unsupervised machine learning method which means it doesn't have labeled data that correspond to observations. Clustering refers to the collection of data points aggregated together because of similarities. The k in K-Means is the target number of centroids which is k , which are the centers of the clusters. After the k number of centroids, the other data points are assigned to the closest cluster while keeping the centroids as small as possible.

Topic Modeling: This is a form of unsupervised machine learning method in which scans documents to identify the phrases and patterns within the documents to create a collection of word groups and expressions that are representative of the documents. Topic modeling is known as picking the best words that represent the subjects from documents. There are various types of algorithms that can be used for Topic Modeling but the one we used was LDA - Linear Discriminant Analysis. In the LDA model each document is viewed as a mixture of topics that are present in the corpus. The LDA model proposes that each word in the document is attributable to one of the document's topics. The LDA model discovers the different topics the document represents and how much of a topic is present in the document.

Sentiment Analysis: Sentiment Analysis is a type of natural language pro-

cessing that is used to create quantitative feedback on qualitative items. This is done through a text analysis method that detects polarity which is either a positive or negative opinion within text. The goal of sentiment analysis is to measure the attitude, sentiment and emotions based on the computational treatment of subjectivity in a text. This is then used to create a net positive, negative, or neutral category for each document based on the classification of the items in that document.

4 Analysis and Discussion

4.1 Bag of Words

The Bag of Words procedure was completed with a term-document matrix using the python genism package, in which a data frame with 415,372 rows and 79 columns was created. In each cell there are two numbers – the first is the term id and the second is the frequency of that term in the document. A representation of the frequency distribution of the twenty most common tokens was also created to supplement the bag of words term-document matrix. This was done using the seaborn plotting using Pandas attributes and x-tick rotation for ease of viewing (Figure 1). This shows a couple of common topics, some of which are expected such as the name of the court case or topic (roevswade, roevwade, abortion, scotus). Alternatively, some give more insight on what this ruling impacts, such as rights, kansas, choice, abortionishealthcare, state, and women.

4.2 TF-IDF

From the sklearn package in Python, using the Vectorizer and tfidfvectorizer which converts a collection of raw documents to a matrix of tf-idf features. The word cloud produced represents the top features from the tf-idf matrix around the tweets from Twitter around the reversal of Roe v Wade (Figure 2). As we can see there are terms around voting, such as November for midterm elections, vote, Texas GOP, democrat, then the supreme court with amendment, supreme, overturning, and then healthcare-related terms such as access, and reproductive health.

This word cloud below can be viewed as a summary of the most important features, representing all of the tweets from the day Roe v Wade was overturned. This showcases some of the most relevant words and topics included in the tweets that day of 415,000. Compared to the frequency plot above from Bag of Words, we can see that while there is a large overlap in terms and frequencies, there are terms that appear in the Bag of Words frequency but not IDF (i.e. Kansas), and terms that appear in TF-IDF but not Bag of Words (i.e. Texas).

TF-IDF is weighted to adjust for inverse document frequency, so the TF-IDF results produced a less biased result. It is apparent that the IDF accounted for roevwade and pulled out more relevant terms that weren't overwhelming.

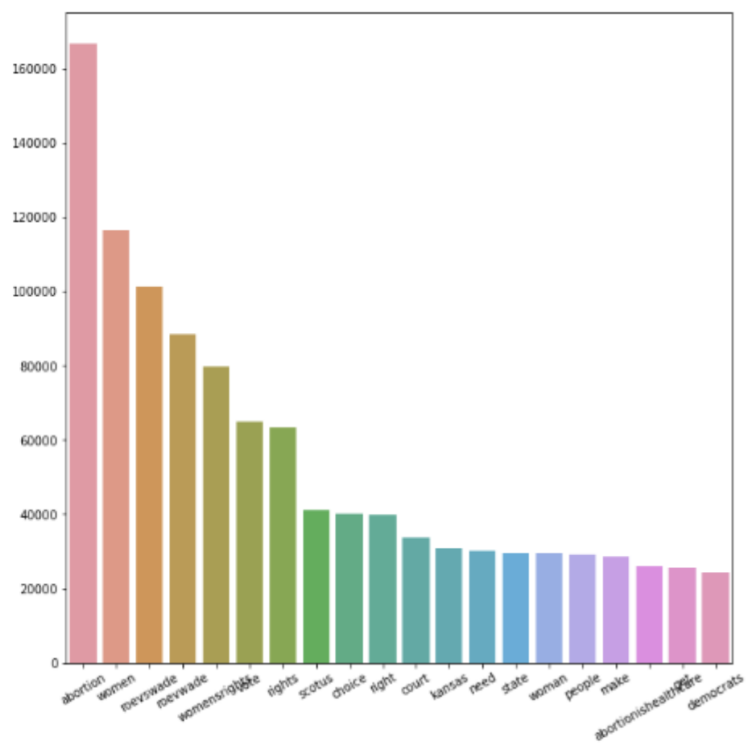


Figure 1: Frequency Distribution of Common Tokens in documents from TEXT



Figure 2: TF-IDF Word Cloud from documents from TEXT

The relevant words it pulled out were around voting, rights, choice, democrat, reproductive health with some showing pro, anti, law and access. The size of these words account for the relevancy in tf-idf. This shows as an overall summary, where a user can understand the bigger picture without having the roe v wade in the center.

4.3 K-Means Clustering

The elbow method looks at the WSS (within the cluster sum of squares) and measures the compactness of the clustering which we want to be as small as possible. The elbow method looks at the total WSS as a function of the number of clusters. Based on the elbow plot, it is apparent that the optimal number of clusters is within the range of 2 to 5 clusters for k-means (Figure 3). While topic modeling and K-means clustering use different algorithms, a similar number of topics can be assumed in Topic Modelling based on these results.

4.4 Topic Modeling

Topic modeling was completed on the preprocessed TEXT data term-document matrix using the LDA model from the `gensim` python package with four topics and five passes. The decision to use 4 topics was built from the K-means clustering analysis, which showed the likelihood of around four groupings of the text.

From this, the following keywords with associated coefficients were derived from the four topics:

Topic 1: woman (0.034), women (0.029), vote (0.025), choice (0.019), right (0.018)

This first topic addresses discussions around women's rights on choosing an abortion. The original *Roe v. Wade* decision was made on the basis that

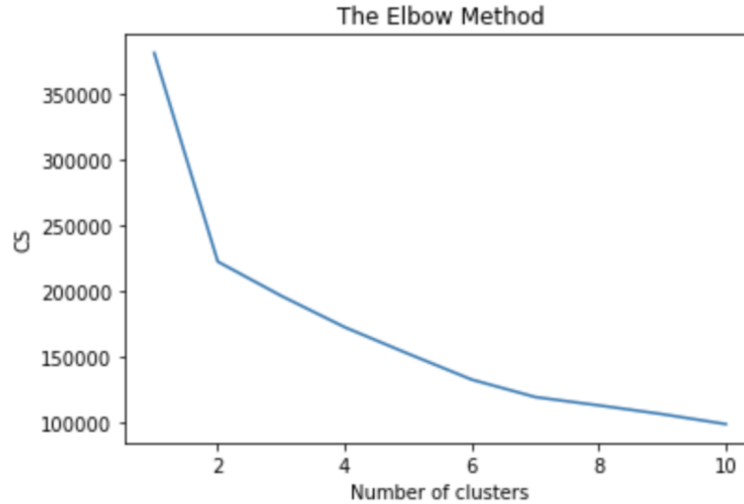


Figure 3: Elbow Method for K-Means Clustering of TEXT documents

abortion access was a right in the United States due to the constitutional right to privacy. As a result, this is a frequently discussed topic regarding the overturn of Roe v. Wade.

Topic 2: adulthumanfemale (0.053), women (0.029), womensrights (0.026), female (0.016), iran (0.014)

This second topic encompasses discussions about the status of women’s rights, specifically how they compare to other countries and the way women are viewed by the United States as a result of Roe v. Wade. The term “adult human female” is used to emphasize the way in which a post - Roe v. Wade United States views women as their ability to have babies. The term Iran is likely included as an example of a country whose government offers women limited rights, and comparing their view on women’s rights with the post Roe v. Wade rights women are given in post Roe v. Wade.

Topic 3: abortion (0.041), roevwade (0.030), state (0.015), court (0.016), scotus (0.012)

The third topic is focused on the legal repercussions of the overturn of Roe v. Wade. This overturn does not ban abortions across the United States, but instead leaves it to the decision of individual states to decide it’s legality. Access to abortion will be different in each state due to laws, constitutions, and court decisions in each state.

Topic 4:womensrights (0.021), women (0.016), abortion (0.012), people (0.010),

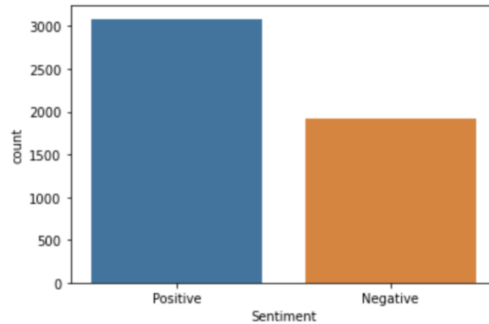


Figure 4: Sentiment Analysis Results by Outcome and Frequency

abortionishealthcare(0.009) This topic is focused on abortion as a form of health-care. Abortion was consistently practiced across the United States as a valid and legal form of health care since the original Roe v. Wade decision, and the result of this overturn is that abortion can no longer be practiced as health care in many states. As a result, people are not able to get the health care that has been a right in the past.

4.5 Sentiment Analysis

Sentiment Analysis was completed via the SentimentIntensityAnalyzer package in python, which uses Vader in the Python nltk package. This analysis was done using the Vader model, which stands for Valence Aware Dictionary for Sentiment Reasoning. This is a model for text sentiment analysis that is sensitive to polarity and the strength of the emotion. From this sentiment analysis, we can determine that the TEXT data from our dataset includes more positive than negative sentiment overall (Figure 4).

5 Conclusion

This analysis of Twitter data was able to give us a better insight and understanding as to the topics and sentiment being discussed on Twitter after Roe v. Wade overturned. The Bag of Words and TF-IDF models presented a high-level overview of the topics being discussed and provided insight on what people were interested in discussing after the overturn. The K-means clustering highlighted that these discussions could be grouped into different categories, and topic modeling is then done to show these categories. The sentiment analysis was then able to give further interpretation to those tweets. Overall, this was an effective way to decipher information from a large dataset of text data.

Overall, we were able to see that there were a couple of topics generally discussed on Twitter after the Roe v. Wade overturn, such as abortion as a women's right, the implication of this decision on women's rights in the United States, the legal implication, and abortion as healthcare. The topics around abortion as women's rights, the implication of the decision on women's rights in the United States, and abortion as healthcare appear to be expressing negative implications from our point of view, and we would have interpreted legal implications as ranging from positive to neutral. However, the sentiment analysis showed overall more positive than negative reactions. For future analyses, it will be important to look further into if this sentiment analysis is inaccurately classifying positive and negative sentiment around the Roe v. Wade discussion, or if more positive sentiment is not well clustered or modeled topically in this analysis. From this analysis alone, we can only conclude these topics generally and that there was overall more negative sentiment noted.

This analysis is limited due to the text-only analysis. We did not remove hashtags as part of this analysis, which could potentially bias this towards certain phrases that are either made to self-summarize the topics discussed or due to certain trends occurring during that time period. Additionally, we did not include any analyses about the influence or interaction of these tweets by incorporating other features into this analysis, such as retweets or followers.

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